Chick Weight Analysis – Summary

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Introduction

Dataset Structure:

- 578 raw observations, 220 for Diet 1 and 120 for Diet 2
- 4 variables: Weight, Time, Chick ID, Diet Type (1-4)
- 50 chicks in total, mostly measured over 21 days

Weight	*Time*	*Chick ID*	*Diet type*
42	*0*	*1*	*1*
51	*2*	*1*	*1*
⋮	⋮	⋮	⋮
205	*21*	*1*	*1*
40	*0*	*2*	*1*
49	*2*	*2*	*1*
⋮	⋮	⋮	⋮
264	*21*	*50*	*4*

Objective:

Determine whether diet 1 gives a significantly different growth than diet 2

Data - Cleaning

Dataset cleaning:

- Most chicks observed across 12 time-steps (day 0, 2, 4 .. 20, 21)
- 5 chicks with incomplete observations (<12) excluded from Complete-Case analysis
- Clean dataset: 45 chicks, each with exactly 12 time-steps observations. **Diet 1**: 16, **Diet 2**: 10, Diet 3: 10, Diet 4: 9

Potential selection Bias:

- 4 out 5 chicks with incomplete observations belongs to Diet 1
- Incomplete chicks from Diet 1 showcased weights **27-56%** below group mean at last timestep of observation;

Data - Exploratory Analysis

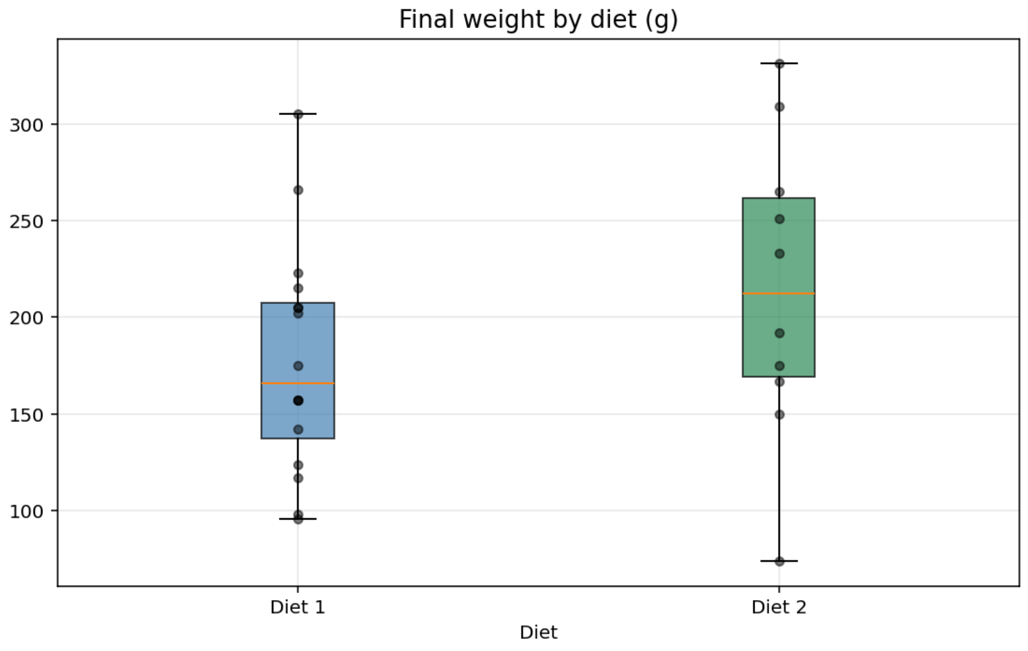
Statistics (final weight) per diet type:

Diet 1

- Mean = 177.75
- Std = 58.70
- Min = 96
- Max = 305
- Nobs = 16

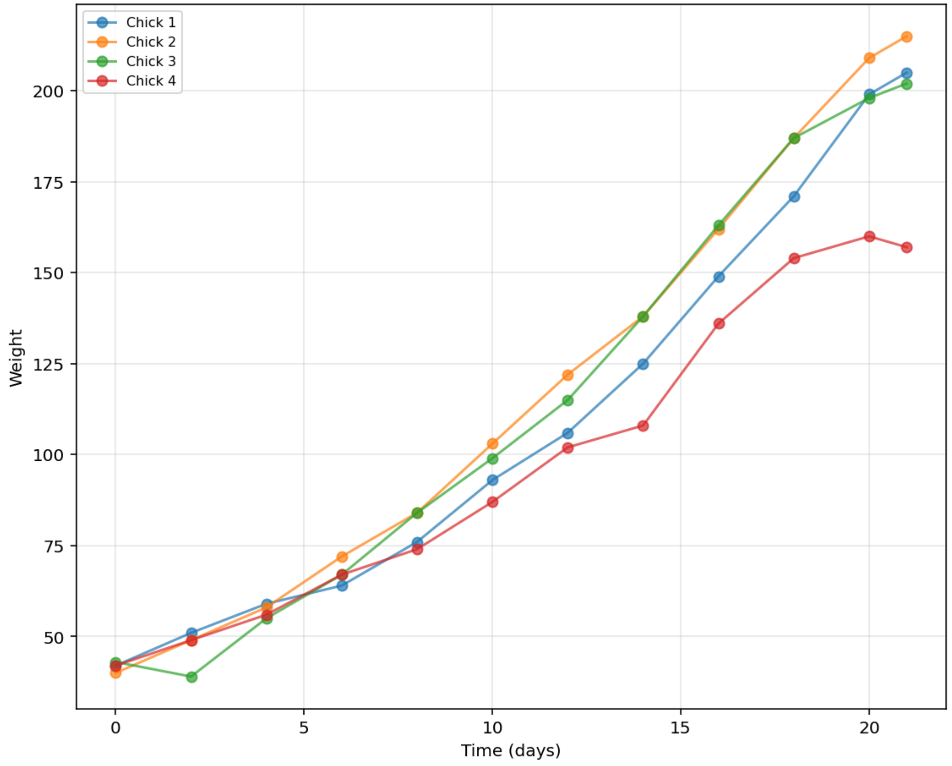
Diet 2

- Mean = 214.70
- Std = 78.14
- Min = 74
- Max = 331
- Nobs = 10

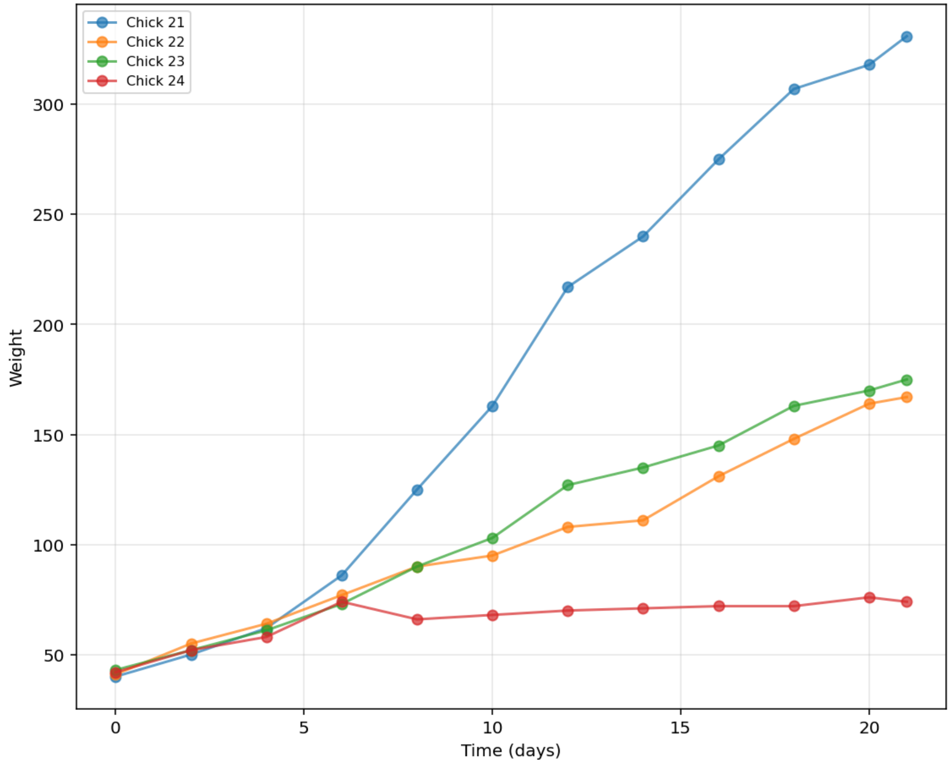


Data - Exploratory Analysis

Growth Trajectories – Diet 1



Growth Trajectories – Diet 2



Both Growth Trajectories justify the use of a linear model with random slope

Methodology

Pre-specified analysis plan, on null-hypothesis H_o: "Diet 1 does not give significantly different growth than Diet 2"

Primary

Linear Mixed Effects Model with Diet Interaction (random slope + random intercept); regression equation:

$$w_{it} = \beta_0 + D_i \cdot \beta_{diet} + t \cdot \beta_{time} + (D_i \cdot t) \cdot \beta_{int} + u_i + t \cdot v_i + \varepsilon_{it}$$

Test H0: $\beta_{int} = 0$, at $\alpha = 0.05$ two-sided

Secondary

- Permutation test, empirical null distribution construction. Regressor:

$$\Delta w = \frac{w_{last\ day} - w_0}{n_{(days)}}$$

<u>Note</u>*: secondary method is* ***corroborative*** *not statistically independent*

Methodology

Pre-specified analysis plan

Primary – Sensitivity on Effect

- Available-LME (n° chicks=30); robustness check to information drop-out in diet 1
- Welch t-test for power analysis

Primary – LME specification & assumptions

Linear Mixed Effects Model

$$w_{it} = \beta_0 + D_i \cdot \beta_{diet} + t \cdot \beta_{time} + (D_i \cdot t) \cdot \beta_{int} + u_i + t \cdot v_i + \varepsilon_{it}$$

Parameters

- β_{diet}: level shift between diets at t=0
- $D_i \in \{0, 1\}$: dummy for diet 2
- β_{time}: average growth-rate for diet 1
- β_{int}: **differential growth-rate** between Diet 1 and 2, primary parameter

Random Effects

- u_i: random intercept of i-th chick
- v_i: random slope of i-th chick
- $(u_i , v_i)^T \sim N(0 , \Sigma)$
- $\Sigma = \begin{pmatrix} \tau_u^2 & \rho\tau_u\tau_v \\ \rho\tau_u\tau_v & \tau_v^2 \end{pmatrix}$, Covariance Matrix

Assumptions

- $\varepsilon_{it} \sim N(0 , \sigma^2)$, residuals are i.i.d
- $(u_i , v_i)^T \perp \varepsilon_{it}$

Primary – LME Results (complete case, n_{chicks}=26)

Parameter	Estimate	SE	z	p-value
Intercept	+31.79	3.47	9.16	<0.001
Time	+6.83	0.94	7.29	<0.001
Diet 2 vs 1 at t=0	-3.84	5.51	-0.70	0.573
Differential Growth Rate (β_{int})	**+1.66 (g/day)**	**1.55**	**1.07**	**0.285**

Confidence Interval on differential growth rate: **[-1.38, +4.70]** g/day at 95%; fail to reject H0 at a=0.05 (p=0.285)

Primary – LME fit visualization

LME predicted growth trajectories with 95% confidence bands

Prediction, Diet 1:

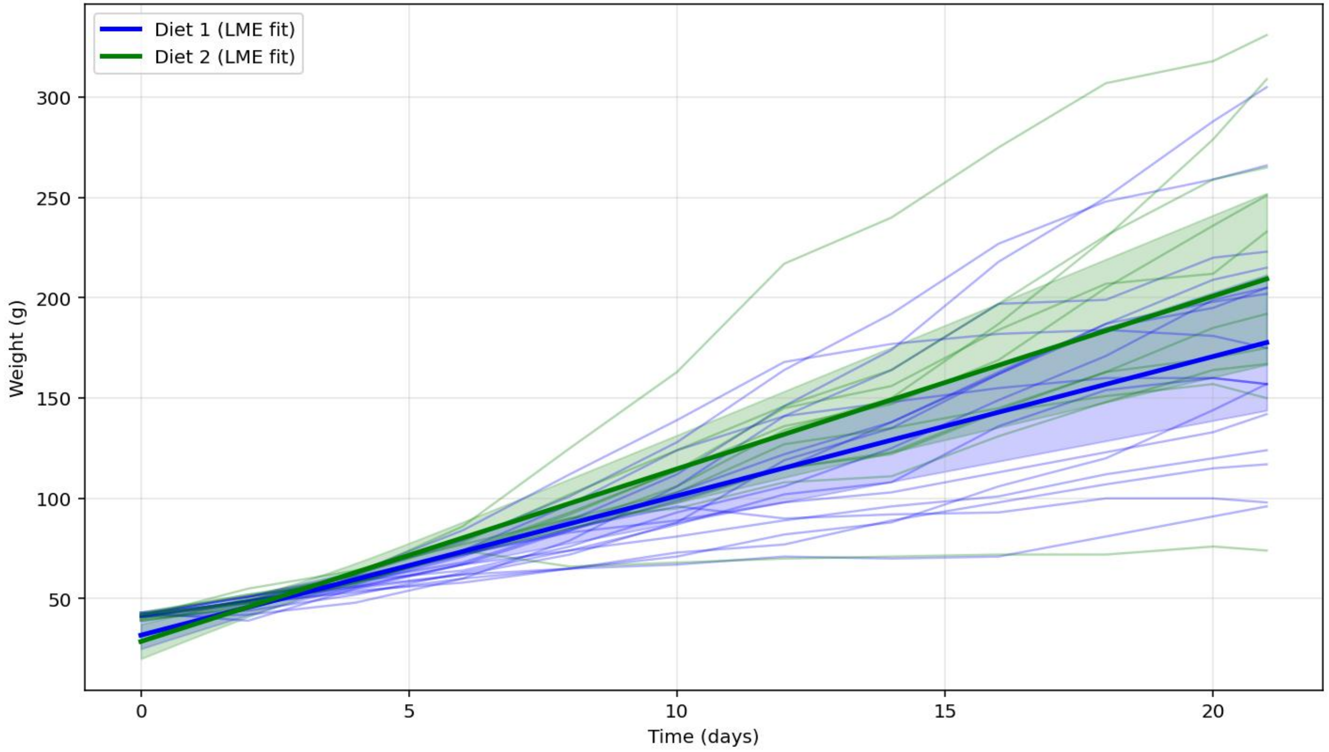
- $\widehat{w}(t) = \beta_0 + \beta_{time} \cdot t$

Prediction, Diet 2:

- $\widehat{w}(t) = \beta_0 + (\beta_{time} + \beta_{int}) \cdot t$

Confidence Bands

- $CB = \pm 1.96 \cdot \sqrt{\mathrm{Var}(\widehat{w}(t))}$



Secondary – Non-Parametric test specifications & assumptions

Resampling via Permutation:

- Randomly permutes diet 1 & 2 labels across 26 chicks, recomputes mean growth-rate difference 10000 times to build null hypothesis distribution

Assumptions:

- Diet labels are exchangeable under H_0 true; no distributional assumptions

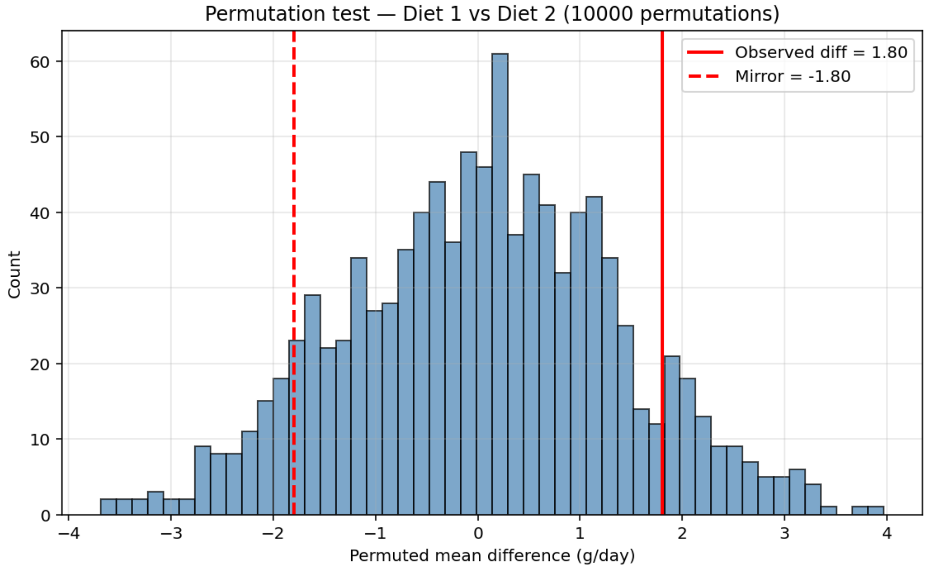
Parameters

- p-value = $\frac{k}{B}$
- $|\Delta w_{obs}|$ = average growth-rate difference observed in data
- k = n° of permutations where $|\Delta w| \geq |\Delta w_{obs}|$
- B = total n° of permutations (10000)

Note: method requires longitudinal data to collapse into a scalar

Secondary – Non-Parametric tests results

Test	Estimate	p-value
Permutation ($n_{iter} = 10000$)	$\lvert \Delta w_{obs} \rvert$ = +1.80 g/day	**0.175**



Result consistent with primary analysis (p=0.285); fail to reject H_0

Sensitivity – Primary, LME with available dataset (n_{chicks}=30)

Parameter	Estimate	SE	z	p-value
Intercept	+33.70	2.94	11.47	<0.001
Time	+6.26	0.82	7.69	<0.001
Diet 2 vs 1 at t=0	-5.06	5.05	-1.00	0.316
Differential Growth Rate (β_{int})	**+2.53 (g/day)**	**1.40**	**1.68**	**0.093**

Confidence Interval on differential growth rate: **[-0.39, +5.09]** g/day at 95%; fail to reject H_o at a=0.05 (p=0.093)

Welch t-test & Power Analysis

Welch t-test results:

- Computed on growth-rate, outputs t-stat **= -**1.304 and p-value = 0.212, not significant

Power Analysis scope

- Computing probability (power) of detecting a true effect of observed magnitude (d=0.563) given the available sample size (based on Welch)
- $n_{80\%}$: number of chicks per group required to gain 80% power

Power Analysis Results

Current Power	27%
$n_{80\%}$	51

Conclusions

Primary method Results

- LME *time x diet* interaction: β_{int} =+1.66 g/day, p = 0.285, 95% CI [-1.38, +4.70] g/day; **fail to reject H_0:** ''Diet 1 does NOT gives a significantly different growth than Diet 2'' at *a*=0.05

Secondary method Results

- Permutation test p-value = 0.175; distributional robustness confirmed, corroborative method

Sensitivity

- (Primary) Available-case LME: p = **0.093** vs Complete-case p=**0.285**; borderline shift consistent with informative dropout in Diet 1 (4 incomplete chicks, 3 out 4 below-average growers), further confirmed by β_{int} =+1.66 g/day from complete case vs +2.53 g/day

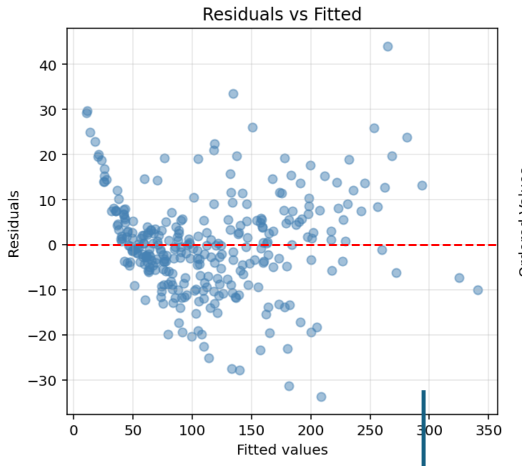
- Confidence Interval from Available-case is more <u>right-skewed</u> (around $\beta_{int} = 0$) than Complete-case analysis, [-1.38, +4.70] -> [-0.39, +5.09] further confirming directional trend (Diet 2 outputting heavier chicks than Diet 1) but still with not enough statistical evidence

Power

- Current power = 27%; non-rejection reflects insufficient sample size, not evidence of equivalence (around 51 chicks per group required for 80% power)

APPENDIX

Heteroskedasticity



LME with Sandwich Robust SE (complete case, n chicks=26)

- βint: +1.661 g/day
- Robust SE: 1.468 (analytic: 1.552)
- 95% CI: [-1.216, +4.537] g/day
- p-value: 0.2579 (analytic: 0.2847)

LME with Sandwich Robust SE (available case, n chicks=30)

- βint: +1.899 g/day
- Robust SE: 1.416
- 95% CI: [-0.877, +4.675] g/day
- p-value: 0.1800